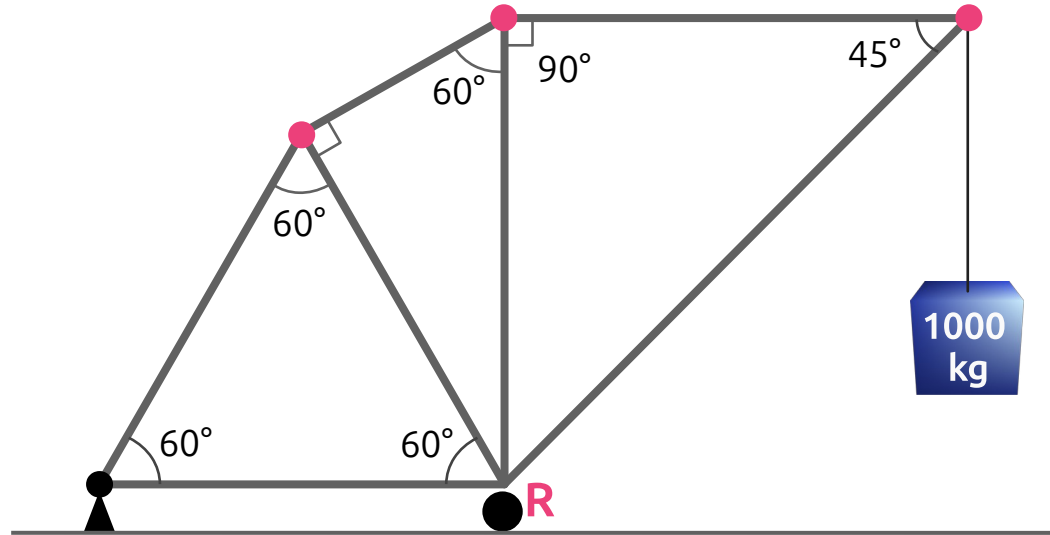


E7 Introduction to Computer Programming for Scientists and Engineers

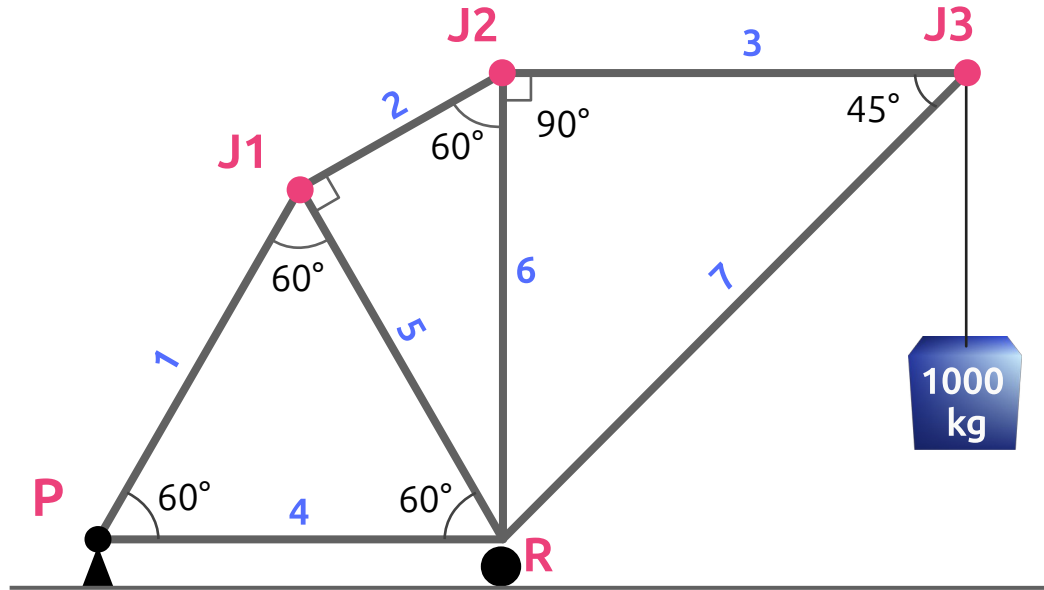
Discussion week 13

**Using linear algebra to
solve the forces in a crane**

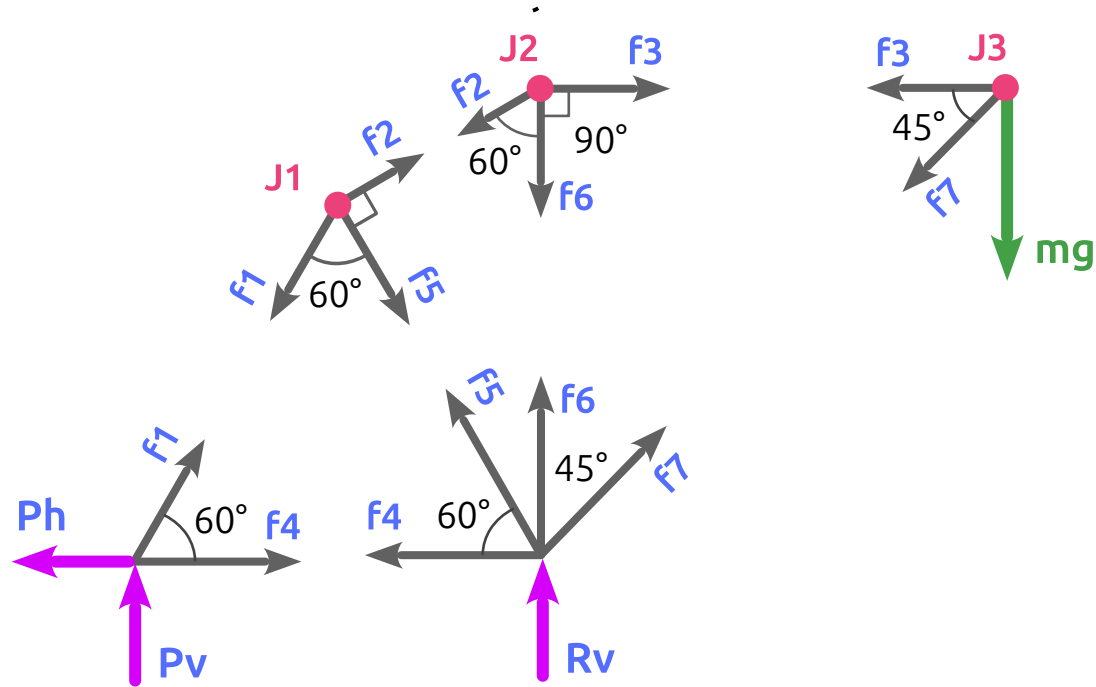
Given the crane shown below, compute the forces present in each of its segments when it lifts a box weighing 1000kg.



Step 1: Label the links and joints.

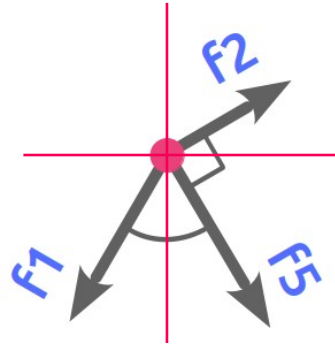


Step 2: Free body diagram for each joint



Step 3: Vertical and horizontal force balance equations

Example **J1**:



System of linear equations

$$J1(h): \sin(30)*f1 - \cos(30)*f2 - \sin(30)*f5 = 0$$

$$J1(v): \cos(30)*f1 - \sin(30)*f2 + \cos(30)*f5 = 0$$

$$J2(h): \cos(30)*f2 - f3 = 0$$

$$J2(v): \sin(30)*f2 + f6 = 0$$

$$J3(h): f3 + \cos(45)*f7 = 0$$

$$J3(v): -\sin(45)*f7 = mg$$

$$P(h): -\sin(30)*f1 - f4 + Ph = 0$$

$$P(v): -\cos(30)*f1 + Pv = 0$$

$$R(h): f4 + \sin(30)*f5 - \cos(45)*f7 = 0$$

$$R(v): -\cos(30)*f5 - f6 - \sin(45)*f7 + Rv = 0$$

Step 4: Put the system of equations into matrix form.

0

J1 (h):

1

J1 (v):

2

J2 (h):

3

J2 (v):

4

J3 (h):

5

J3 (v):

6

P (h):

7

P (v):

8

R (h):

9

R (v):

0

f1

1

f2

2

f3

3

f4

4

f5

5

f6

6

f7

7

Ph

8

Pv

9

Rv

s30	-c30			-s30					
c30	-s30			c30					
	c30	-1							
	s30				1				
		1				c45			
						-s45			
-s30			-1				1		
-c30								1	
			1	s30		-c45			
				-c30	-1	-s45			1

f1

f2

f3

f4

f5

f6

f7

Ph

Pv

Rv

mg

Step 5: Solve in code.

